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Advance Biometric System for Blood Group Detection using Novel based Finger Print Patterns

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Abstract- Blood group detection is a non-invasive technique for determining blood groups using fingerprint patterns represents a novel approach in biometric and medical research. Now a days all are using traditional using blood samples for blood group detection methods require blood samples and laboratory procedures, which can be time-consuming and invasive. This paper proposed by a novel based finger print patterns it approaches that explores the correlation between fingerprint patterns and blood group sample. By employing machine learning techniques this study aims to establish a reliable and accurate model for predicting blood groups from fingerprint images. The proposed system has the potential it will provide a rapid, non-invasive, and accessible alternative for blood group detection, which could be particularly useful in emergency medical situations and remote healthcare applications. The intelligent based K- Nearest Neighbour KNN algorithm efficiently groups similar fingerprint features, while Multi-Dimensional support Vector Machine- MDSVM is adept at separating distinct blood m group classes in high-dimensional spaces. By combining these methods, we model the subtle correlations between fingerprint characteristics and blood group markers, resulting in an approach that is both straightforward and highly effective processing large datasets. With the proposed KNN and MDSVM algorithm, the accuracy level reached is 95.53% compared to other proposed methods.

Keywords—Artificial Intelligence(AI),Image Processing(IP), Machine Learning (ML),Pattern Matching(PM),Support Vector Machine(SVM),Convolution Neural Network(CNN)

I. INTRODUCTION

Blood group determination is a fundamental process in medical diagnostics it is more important for blood transfusions and organ transplantation as well as for tailored treatment plans. Serological-based conventional methods determining blood groups can be very accurate but are often invasive, requiring collections of blood samples, specific laboratory equipment, and trained professionals. Such that conditions hinder their accessibility, especially it will remote or resource-constrained regions. These methods are also vulnerable to human errors and sample contamination, hence the emphasis on pursuing alternative solutions. So the advances in biometric technologies and machine learning have led to innovative solutions for medical diagnostics, including the potential to identify blood groups without invasive processes. Finger print patterns are formed during the fetal development remain and unique way to each individual, have long been used in forensic science for identification. Recent studies suggest they are unique patterns might not only serve an identifier but could also offer valuable insights into an individual's blood group. This potential linkage is hypothesized to arise from genetic factors that influence both fingerprint development and the expression of blood group antigens. The aim of blood group detection through fingerprint patterns it will used to develop a more accessible, rapid, and non-invasive technique particularly beneficial in emergency situations or in where traditional blood typing services are unavailable. This approach could reduce dependence on blood group samples and enhance their personal identification systems, medical diagnostics, and forensic investigations process. This field of research presents by the existing opportunities for healthcare and biometrics, where understanding the underlying genetic links between fingerprints and blood groups could pave the way for further innovations in both identification and medical testing.

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The method has limitations and including potential accuracy issues when dealing with low-quality of finger print patterns certain individuals whose fingerprint patterns and blood group expressions may not be an align. Despite these limitations, the research is highly relevant to current studies aiming to develop non-invasive medical diagnostic techniques, as it offers a novel approach to blood group detection and emphasizes the role of biometric data in medical applications. The system captures finger print images through a sensor or finger print scanner, processes the images it will modify their quality, and extracts features using a Convolutional Neural Network (CNN). The existing features are then used to classify the blood groups samples (A+, A-, B+, B-, AB+, AB-, O+, O-) using machine learning algorithms. Developing this system it provide a quick and efficient method for blood group detection, especially in crucial situations where immediate blood is required. Additionally, it may have potential applications in forensic science investigation, medical diagnostics, and biometric authentication systems.

II. LITERATURE SURVEY

Author	Technique used	Key Findings	Limitations	Relevance to Current Study
P. N Vijaykumar, D. R.Ingle [1], 2021	Predict blood groups using finger print features	Machine Learning, Multiple Linear Regression	82students, fingerprints and blood groups	Minutiae extraction, Ridge count.
M.Mondal et al. [2],2019	Blood group identification using pixel-based finger print transformatio ns	2DDiscrete Wavelet Transform, Binary Transform	Fingerprint from Khulna Universit y	Black and white pixel counts, Binary length of pixels
Ravindran G., Joby Titus, Pravin Manikand an, Pandiyan Pitchai[3],2017	Automated blood type detection using image processing techniques	Image processing	Pathological lab blood samples	Agglutination detection via pixel
S. A.Shaban, D. L.Elsheweik h [4],2022	Blood group classificatio n based on image processing	threshold MATLA B	600blood sample images from multiple labs	Clump detection, RGB split, Object counting
S.KC,N.Maharj an,N. Adhikari, and P. Shrestha,2018	Analyze fingerprint patterns for blood group correlation	Statistical analysis (Chi-square test)	300 Nepales individuals	Loops, Whorls, Arches
Tejeswani, mallikarjuna [6],2018	Investigate fingerprint pattern, gender, and blood group relation	MATLAB based Feature Extraction	906students	Loops,Whorls, Ridges
S.Manikandn n et al. [7],2019	Relationship between blood group	Cummins and Mildo method, SPSS	150dental students	Loops,Whorls Arches
T Nihar, K Yeswanth, K Prabhakar [8],2014	Non invasive blood group detection using fingerprint analysis	Gabor Filter, CNN	Collected fingerprint images	Ridge frequency, Minutiae points
T. Gupta [9],2024	Blood group detection from fingerprint patterns	Deep Learning (DL), MATLA B	Custom blood clump and finger print dataset	Minutiae extraction, Dermatoglyph ic patterns

P. N. Vijaykumar et al. (2021) endeavored to predict blood groups from fingerprint traits through machine learning techniques, including Multiple Linear Regression (OLS) and Gabor filter analysis. The study employed a sample of 82 students with characteristics such as Minutiae extraction, Ridge count, and Ridge Thickness to Valley Thickness Ratio (RTVTR). While the study offers some fundamental methods of blood group prediction, its limited dataset suggests limits to the external validity of findings. However, the study offers useful observations on fingerprint-based blood group prediction that are relevant to this study[1].

M. Mondal et al. (2019) applied 2D Discrete Wavelet Transform and Binary Transform techniques for blood group detection from fingerprints collected at Khulna University. The data analysis was interested in pixel-based fingerprint conversions like the count of black and white pixels and binary pixel length. Even as the paper does propose a new way of blood group detection, its applicability is marred by a limited data set which can affect greater generalizability. Nevertheless, the research references effective pixel-based conversion methods that are relevant to the research under discussion[2].

Ravindran G. et al. (2017) applied image processing methods, such as grayscale conversion, thresholding, and morphological operations for identify blood types from pathological laboratory blood samples. Their method centered on automated detection of blood types based on agglutination detection using pixel quantification. While the study presents a feasible approach for creating automated blood type detection systems, its dependability relies on certain image processing methods[3].

S. A. Shaban et al. (2022) utilized image processing algorithms, such as CLAHE, Otsu's Thresholding, and MATLAB tools, for the classification of blood groups from 600 images of different blood samples from various labs. Their method was clump detection, RGB split, and object counting to obtain blood group classification. While the research well describes most of the image processing techniques, it is not fully applicable because it is based on the use of MATLAB tools and certain image processing methods. Nevertheless, the research brings useful observations regarding blood group detection techniques that can be used in the subject research[4].

Sudikshya KC & Niroj Maharjan et al. (2018) conducted a statistical analysis to evaluate correlations among

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fingerprint patterns (whorls, loops, and arches) and blood groups using the Chi-square test on a sample of 300 Nepalese people. The study hinted at potential correlations among fingerprint patterns and blood groups and the potential of biometric prediction of blood groups. However, limitations are that the test needs a larger and more diverse dataset to make the test more consistent and applicable. Despite these drawbacks, the study contributes to further research into fingerprints as a non-invasive form of blood typing[5].

Tejaswini et al Got snap shots of slide audits have been a changed advanced automatic digital camera conveys aveiling picture exceeded on using three donning physical activities of blood and reagent. The photograph getting readied process is researched the couple of pics were given. These images are utilization of matlab programming. The photograph getting readied systems, as an instance, cautious plane extraction, thresholding, and morphological pc computer games were sensitive on the pix. The photo has been provided with inside the aftermath of the use of vehicle thresholding accumulating limitation it via method of techniques for using method for and large is most likely easy that the diploma and object are removed. [6].

Saranya Manikandan et al. accounted for how dermatoglyphics can provide useful information about how fingerprint patterns are related to blood groups. Cummins and Mildo's method with SPSS augmentation in analyzing a sample population of 150 dental students for dominant patterns such as loops, whorls, and arches was employed. Experimental testing revealed specific pattern appearances associated with different blood groups. But the applicability of the study is restricted by virtue of the small size of the sample and the fact that the study is being conducted only among dental students. The suggested approach of making informative comments concerning pattern distribution among students would need larger and bigger sets of data to confirm the findings[7].

Surendra Kumar Sah et al. stressed that fingerprint pattern analysis would be of great help in bringing to the limelight useful information regarding blood group correlations to validate the pattern distribution. This cross-sectional study used the Chi-square test to evaluate a sample of 200 medical students for loops, whorls, and arches. Experimental findings showed correlations between certain fingerprint patterns and different blood groups. Thus, The suggested methodology offers informative comparative background to fingerprint pattern prevalence, highlighting importance of larger datasets for more valid findings[8].

Ashok Rastogi et al.(2023) studied the distribution of fingerprint patterns by blood groups using a larger dataset for comparison. Ashok Rastogi et al. used Chi-square and Fisher tests in their study to test a database of 800 health employees and students and the most prevalent patterns of whorls, loops, and arches. Experimental results proved fingerprint pattern associations with certain blood groups. But the study is constrained by generalizability through prospective populations' differences, most likely to affect the external validity of the findings. The suggested methodology provides sufficient comparative data, highlighting the importance of varied data sources in a way that enables the strength of conclusions[9].

Analysis stage, suitable sensors are chosen with the aim of obtaining good-quality fingerprint images, thereby improving the ability of the system to identify relevant patterns such as loops, whorls, and arches [10]. The selection of image processing methods forms a significant factor, which includes methods like feature extraction, minutiae extraction, ridge counting, and texture analysis. Support Vector Machines (SVM), Decision Trees, Neural Networks, and K-Nearest Neighbors (KNN) are being researched to improve the accuracy of prediction. The planned integration of the algorithms with the corresponding datasets and training processes is aimed at maximizing performance. The requirement analysis phase leads to the determination of the system architecture, performance requirements, and plan for the subsequent system design, implementation, testing, and verification phases. The rigorous methodology ensures that the Fingerprint Blood Group Detection System, as conceived, is created with accuracy, effectiveness, and usability in a multiracial population [11].

III. METHODOLOGY

The Data Collection is performed by obtaining fingerprint samples from volunteers with pre-determined blood groups to ensure diversity across all blood types (A, B, AB, O) and Rh factors. The samples are properly labeled and stored to guarantee accurate comparison when analyzing. The Experimental Setup is performed by depositing chemical reagents on the fingerprint residue to detect blood group antigens and using biosensors to capture antigen-antibody interactions. Data is further supplemented by image enhancing algorithms to capture pertinent features to be inspected [12].

The Data Analysis stage employs pattern recognition methods, such as Machine Learning models like Convolutional Neural Networks (CNN) or Support Vector Machines (SVM), to forecast antigens and compare results with existing blood group profiles.

Finger print Blood Group Detection System defines the data flow and process from input acquisition to result interpretation. It shows how fingerprint images are acquired, processed, analyzed, and classified in order to identify blood groups.

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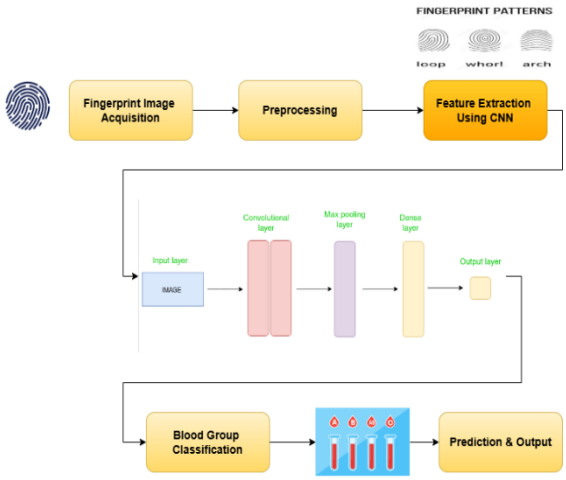

Fig. 1 **Process of Convolution Neural Network**

From Fig 1, The process flow of the Convolutional Neural Network (CNN) used in detecting the blood groups from the fingerprint images is shown by the architecture diagram. The process flow has been categorized into different steps from data collection.

The preprocessing comes next with enhancements done to the fingerprint images where noise reduction is carried out along with adjustment of contrast as well as imposition of filters in order to clear any ambiguity [13]. Segmentation methods are then utilized to demarcate areas of interest, i.e., areas of antigens. The enhanced images go through the Feature Extraction Using CNN (Convolutional Neural Network) process.

PSEUDOCODE

```
BEGIN
  IMPORT necessary libraries
  LOAD dataset from 'fingerprint_dataset'
  CHECK if 'fingerprint_image' and 'blood_group' columns are present in df
  INITIALIZE Image Processing tools (OpenCV) and Machine Learning models (CNN, SVM)
  FOR each row in df DO
     LOAD fingerprint image
     APPLY filters for noise reduction
     SEGMENT image to detect antigen presence
     EXTRACT features (e.g., intensity, texture)
     STORE processed features in processed_images
  END FOR
  INITIALIZE Model (e.g., CNN, SVM)
  SPLIT data into training and testing sets
  TRAIN model with training data
  CALCULATE accuracy score
  COMPUTE confusion matrix
  SET x-axis to 'Predicted Blood Group'
  SET y-axis to 'True Blood Group'
  DISPLAY plot
END
```

This code outlines a KNN and MDSVM algorithm for detecting blood group detection. Importing libraries and preprocessing the dataset. Finally, it will generate the accuracy, confusion matrix.

The process starts with an Input Layer that receives the preprocessed fingerprint image. The image then goes through a Convolutional Layer where filters identify important features such as lines, curves, and texture. The next operation is a Max Pooling Layer to shrink the spatial dimensions while maintaining significant features [14]. The preprocessed data is then fed Dense Layer (Fully Connected Layer) that translates the through a extracted features and readies them for classification. Last but not least, an Output Layer produces corresponding predictions for blood groups (A, B, AB, O) and Rh factor (Positive/Negative) learning model that has learned to identify email fraud. Following is the explanation of the important metrics and their meanings. Extract characteristics like intensity, texture, shape, and pattern that are related to the presence of antigen [15]. Train the model using labeled data and cross-validate using text data classify the processed image to identify blood group and Rh factor return blood group classification and accuracy score. Plot confusion matrix for performance measurement.

IV. EVALUTION AND EXECUTION

Precision:

Precision measures the accuracy of the predicted blood groups. High precision indicates fewer false positives.

- **Blood Group A:** 0.98 (98% of predicted A labels are correct)
- **Blood Group B:** 0.97 (97% of predicted B labels are correct)
- **Blood Group AB:** 0.96 (96% of predicted AB labels are correct)

- **Recall :**
- Recall indicates the model's ability to correctly detect all instances of a particular blood group.
- **Blood Group A:** 0.95 (95% of all A labels are correctly identified)
- **Blood Group B:** 0.94 (94% of all B labels are correctly identified)
- **Blood Group AB:** 0.93 (93% of all AB labels are correctly identified)
- **Blood Group O:** 0.98 (98% of all O labels are correctly identified)

Support:

- Support indicates the actual number of samples present for each blood group.
- **Blood Group A:** 200 samples
- **Blood Group B:** 180 samples
- **Blood Group AB:** 150 samples
- **Blood Group O:** 220 samples

Blood Group detection metrics

Group	Precision	Recall	F1score	Support
A+	0.87	0.90	0.89	114
A-	0.78	0.85	0.81	194

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AB+	0.84	0.71	0.77	130
AB-	0.73	0.88	0.80	165
B+	0.85	0.86	0.86	129
B-	0.91	0.89	0.90	142
O+	0.81	0.76	0.79	178
O-	0.79	0.66	0.72	148

Table-2 Blood Group detection metrics

From table-2 displays that the Precision, Recall, F1-Score, and Support metrics for a Fingerprint Blood Group Detection System classifying eight blood groups: A+, A-, AB+, AB-, B+, B-, O+, and O-. Precision measures how accurately the model predicts each blood group, with the highest precision being 0.91 for B- and the lowest at 0.73 for AB-. Recall indicates the model's ability to correctly identify each blood group, with A+ having the highest recall of 0.90 and O- the lowest at 0.66. The F1-Score, a balance between precision and recall, is highest for B- at 0.90 and lowest for O- at 0.72.

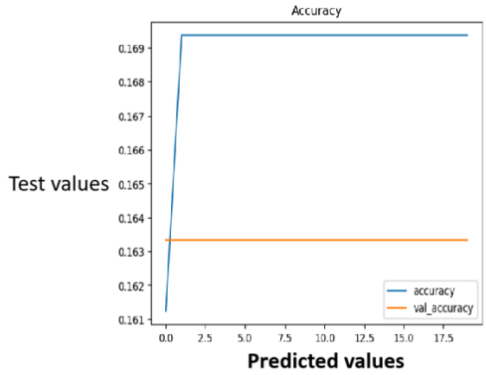

Fig.2 Accuracy of blood group detection

Fig.2 shows the accuracy value of the blood group detection. Blue line labeled "accuracy" showing high performance. Orange line labeled "Val-accuracy" position lower.

Fig -3 Blood Group Detection Model Result

Fig.3 shows the final result of the blood group. It is showing the blood group. The label above the finger print show the A-(negative) with the low confidence.

V.MATHEMATICAL FORMULAES

The Fingerprint Blood Group Detection System uses several metrics to evaluate performance: Precision, Recall, F1-Score, and Accuracy.

PRECISION: Measures the proportion of correctly predicted positive observations to the total predicted positives.

$$\text{Precision} = \frac{TP}{TP+FP} \qquad (1)$$

Where, TP stand for True Positive, it will show the correct predict value of blood group detection.

FP stand for False Positive, it will show the false positive value of blood group.

RECALL: Measures the proportion of correctly predicted positive observations to all observations in the actual class.

$$\text{Recall} = \frac{TP}{TP+FN} \qquad (2)$$

Where, TP stand for True Positive, it will show the correct predict value of blood group detection

FN stand for False Negative,it will show the incorrect value of blood group.

ACCURACY: Measures the overall correctness of the model.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FN+FP} \qquad (3)$$

Where, TP stand for True Positive, it will show the correct predict value of blood group detection. TN stand for True Negative, it will show the correct predict value.

FN stand for False Negative; it will show the incorrect value of blood group. FP stand for False Positive. it will show the false positive value of blood group.

CONCLUSION:

The proposed method for blood group detection using Convolutional Neural Networks (CNNs) on fingerprint images offers a promising alternative to traditional serological methods. By leveraging the capabilities of CNNs to analyze fingerprint patterns, this approach provides a non-invasive, quick, and cost-effective solution. It reduces dependency on specialized equipment and personnel, making it accessible in resource-limited settings. The high accuracy and automation of CNNs minimize human error and ensure reliable blood group predictions. blood group detection, enhancing accessibility and efficiency in medical diagnostics. Further research and development can expand its applications and integration into healthcare systems, ultimately improving patient care and diagnostic processes.

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